# School of Computer Science & Engineering

## COURSE PLAN

## Academic Year 2025-26 EVEN SEMESTER

| | |
|---|---|
| **School/Department of Students** | **School of Computer Science & Engineering** |
| **Name of the Program(s) of Students** | **B.Tech** |
| **PRC Approval Ref. No.** | |
| **Semester/Year** | **IV/II** |
| **Course Code & Name** | **CSE2262** |
| **Credit Structure (L-T-P-C)** | **3-1-0-4** |
| **Contact Hours** | **4** |
| **Course In-Charge (IC)** | **Dr Madhusudhan M V, Dr Aarif Ahamed S, Dr. Praveena K N** |
| **Course Instructor(s)** | **Dr. Madhusudhan M V,Dr. Aarif Ahmed,Dr. Praveena K N,Dr. Manjula H M,Mr. Anandan B,Mr. Asad Mohammed Khan,Ms. Babitha,Mr. Dinesh C,Ms. Chandana,Mr. Girish Kumar,Dr. Jagadevi Bakka,Mr. Jaikumar,Ms. Meena Kumari K S,Mr. Mohammed Shakir,Ms. Padma T G,Dr. Pallavi R,Ms. Sandhya L Mr. Shankar J,Dr. Shanthi S,Mr. Sheikh Wasim Masood Ms. Shet Reshma,Dr. Sridevi S,Ms. Suma N G,Dr. Yamanappa,Dr. Pamela Vinitha,Ms. Anu Joseph Dr. Jayachandran A,Dr. Ruhin Kouser, Mr. Katta Sriram,Ms. Priyanka Gour, Dr. Sandeep Albert Mr. Sakthi S** |
| **Course URL** | **https://presidencyuniversity.linways.com** |

### 1. COURSE PRE-REQUISITES:

*CSE2253 Data Structure.*

### 2. COURSE DESCRIPTION:

This course introduces techniques for the design and analysis of efficient algorithms and methods of applications. This course discusses the classic approaches for algorithm design such as Divide and Conquer, Dynamic Programming, Greedy method. This course also describes other basic strategies searching solution space. The core concepts of analyzing algorithms and classifying them into various complexity classes is covered in the end.

## 3. COURSE OBJECTIVES:

The objective of the course is to familiarize the learners with the concepts of **Analysis of Algorithms** and attain **Skill Development** through **Problem Solving** Methodologies.

## 4. COURSE OUTCOMES:

**TABLE 1: COURSE OUTCOMES**

| CO Number | Statement of CO<br>*On successful completion of the course the students shall be able to* | Blooms Cognitive Level |
|---|---|---|
| CO1 | Analyze the efficiency of a given algorithm | Analyze |
| CO2 | Apply divide & conquer, and decrease & conquer technique for searching and sorting Problems | Apply |
| CO3 | Apply Dynamic Programming & greedy technique for a given problem | Apply |
| CO4 | Apply space & time tradeoff technique and limitation of algorithms for solving a Problem | Apply |
| CO5 | Demonstrate Back tracking technique and branch & bound technique for solving the complex problems | Apply |

## 5. MAPPING OF COURSE OUTCOMES WITH PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

### 5.1 *PROGRAM OUTCOMES*:

On successful completion of the Program, the students will be able to:

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

**TABLE 2a: CO-PO Mapping**

| CO. No | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO1** | H | H | M | L | | | | L | | L | | M |
| **CO2** | H | M | L | | | | | L | | M | | M |
| **CO3** | H | M | L | | | | | L | | M | | M |
| **CO4** | H | M | L | | | | | L | | M | | M |
| **CO5** | H | M | L | | | | | L | | M | | M |

### 5.2 PROGRAM SPECIFIC OUTCOMES:

On successful completion of the Program, the students will be able to:

| | |
|---|---|
| PSO1 | [Problem Analysis]: Identify, formulate, research literature, and analyze complex engineering problems related to Software Engineering principles and practices, |

| | |
|---|---|
| | Programming and Computing technologies reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences. |
| PSO2 | [Design/development of Solutions]: Design solutions for complex engineering problems related to Software Engineering principles and practices, Programming and Computing technologies and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations. |
| PSO3 | [Modern Tool usage] : Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities related to Software Engineering principles and practices, Programming and Computing technologies with an understanding of the limitations. |

| TABLE 2b: CO-PSO Mapping | | | |
|---|---|---|---|
| **CO Number** | **PSO1** | **PSO2** | **PSO3** |
| **CO1** | H | M | |
| **CO2** | M | M | |
| **CO3** | M | M | |
| **CO4** | M | L | |
| **CO5** | M | L | |

## 6. COURSE CONTENT:

| Module Number | Module Name | Number of Sessions |
|---|---|---|
| 1 | **Introduction:** Notion of algorithm, Importance of algorithm, Important Problem Types, **Fundamentals of analysis of algorithm efficiency**: General equation, Order of growth, basic efficiency classes, asymptotic notations, mathematical analysis of recursive & non-recursive algorithms, **Brute Force technique:** Bubble sort, selection sort, linear search, brute force string matching, Exhaustive search: Travelling salesperson problem, knapsack problem | 14 |
| 2 | **Divide & Conquer technique:** General Method, General recursive equation, Masters Theorem, Binary search, Merge sort, quick sort. **Decrease & Conquer technique:** General method, insertion sort, topological sorting | 11 |
| 3 | **Dynamic Programming:** General method, Warshall algorithm, Floyd algorithm, 0/1 knapsack, **Greedy method:** General method, prim's algorithm, Kruskal algorithm, Dijkstra's algorithm, Huffman coding | 13 |
| 4 | **Space and time tradeoff:** introduction, Input enhancement in string matching: Horspool algorithm and Boyer-Moore algorithm, **Limitations of algorithm power:** Introduction, lower bound arguments, decision trees, P, NP and NP-complete problems | 11 |
| 5 | **Coping with limitations of algorithm power:** Introduction, **Backtracking**: n-queens problem, subset-sum problem, Hamiltonian circuit problem, **Branch & bound:** Travelling salesperson problem, knapsack problem, Job assignment problem. | 11 |

***REFERENCE MATERIALS:***

**Text Books:**

**T1:** Anany Levitin, *"Introduction to the Design and Analysis of Algorithms*", 3rd edition, Pearson Education, 2018.

**T2:** Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest and Clifford Stein, *"Introduction to Algorithms*", 4th edition, MIT Press, 2022.

**Reference Books:**

R1. AV Aho, J Hopcroft, JD Ullman, *"The Design and Analysis of Algorithms"*, Addison-Wesley, 1974

R2. Tim Roughgarden, *"Algorithms Illuminated"* (books 1 through 3), "Operating Systems Design and Implementation", Soundlikeyourself Publishing, 2017-2019.

R3. J. Kleinberg and E. Tardos, "*Algorithm Design*", Addison-Wesley, 2005

**Online Resources**

1. EBook: https://presiuniv.knimbus.com/user#/home
2. https://onlinecourses.nptel.ac.in/noc26_cs67/
3. www.youtube.com/watch?v=so4zT-u09LM&list=PL8GCet93Cu1iRF611mrUu-Rhy12fMwZOf

## 7. DETAILED SCHEDULE OF INSTRUCTION

**TABLE 3: LESSON PLAN**

| Session Number | Topic | Sub-Topic | CO Number | Reference |
|---|---|---|---|---|
| **1.** | Program Integration & Course Integration | Overview of the Course, Scopes and Opportunities | | |
| **Module 1** | | | | |
| **2.** | Introduction | Notion of algorithm, Importance of algorithm, | CO1 | T1: Ch. No. 1, pp 3 |
| **3.** | | Important Problem Types | CO1 | T1: Ch. No. 1, pp 19 |
| **4.** | Fundamentals of analysis of algorithm efficiency | General equation, Order of growth | CO1 | T1: Ch. No. 2, pp 41 |
| **5.** | | basic efficiency classes | CO1 | T1: Ch. No. 2, pp 52 |
| **6.** | | asymptotic notations | CO1 | T1: Ch. No. 2, pp 56 |
| **7.** | mathematical analysis of non-recursive algorithms | Array uniqueness problem, finding maximum element | CO1 | T1: Ch. No. 2, pp 61 |
| **8.** | | Counting number of bits required to represent the given decimal number in binary | CO1 | T1: Ch. No. 2, pp 65 |
| **9.** | mathematical analysis of recursive algorithms | factorial | CO1 | T1: Ch. No. 2, pp 78 |
| **10.** | | Counting number of bits required to represent the given decimal number in binary | CO1 | T1: Ch. No. 2, pp 80 |
| **11.** | Brute Force technique | Bubble sort | CO1 | T1: Ch. No. 3, pp 98 |
| **12.** | | selection sort | CO1 | T1: Ch. No. 3, pp 100 |
| **13.** | | linear search | CO1 | T1: Ch. No. 3, pp 103 |

| | | | | |
|---|---|---|---|---|
| **14.** | | brute force string matching | CO1 | T1: Ch. No. 3, pp 104 |
| **15.** | Exhaustive search | Travelling salesperson problem, knapsack problem | CO1 | T1: Ch. No. 3, pp 114 |
| **Module 2** | | | | |
| **16.** | Divide & Conquer Technique | General method, General recursive equation | CO2 | T1: Ch. No. 4, pp 123 |
| **17.** | | Master's theorem, | | T1: Ch. No. 4, pp 124 |
| **18.** | | Binary Search | CO2 | T1: Ch. No. 4, pp 124 |
| **19.** | | Merge sort | CO2 | T1: Ch. No. 4, pp 125 |
| **20.** | | Merge sort | CO2 | T1: Ch. No. 4, pp 125 |
| **21.** | | Quick sort | CO2 | T1: Ch. No. 4, pp 129 |
| **22.** | | Quick sort | CO2 | T1: Ch. No. 4, pp 129 |
| **23.** | | General method | CO2 | T1: Ch. No. 5, pp 157 |
| **24.** | Decrease conquer technique | Insertion sort | CO2 | T1: Ch. No. 5, pp 160 |
| **25.** | | Topological sorting | CO2 | T1: Ch. No. 5, pp 172 |
| **26.** | | Topological sorting | CO2 | T1: Ch. No. 5, pp 172 |
| **Module 3** | | | | |
| **27.** | Dynamic Programming | General method | CO3 | T1: Ch. No. 8, pp 279 |
| **28.** | | Warshall algorithm, | CO3 | T1: Ch. No. 8, pp 284 |
| **29.** | | Floyd algorithm | CO3 | T1: Ch. No. 8, pp 288 |
| **30.** | | Floyd algorithm | CO3 | T1: Ch. No. 8, pp 288 |
| **31.** | | 0/1 knapsack | CO3 | T1: Ch. No. 8, pp 299 |
| **32.** | | 0/1 knapsack | CO3 | T1: Ch. No. 8, pp 299 |
| **33.** | Greedy method | General method | CO3 | T1: Ch. No. 9, pp 307 |
| **34.** | | prim's algorithm, | CO3 | T1: Ch. No. 9, pp 308 |
| **35.** | | Kruskal algorithm | CO3 | T1: Ch. No. 9, pp 315 |
| **36.** | | Dijkstra's algorithm | CO3 | T1: Ch. No. 9, pp 323 |
| **37.** | | Dijkstra's algorithm | CO3 | T1: Ch. No. 9, pp 323 |
| **38.** | | Huffman coding | CO3 | T1: Ch. No. 9, pp 328 |
| **39.** | | Huffman coding | CO3 | T1: Ch. No. 9, pp 328 |

| | | | | |
|---|---|---|---|---|
| **Module 4** | | | | |
| **40.** | Space and time tradeoff | introduction | CO4 | T1: Ch. No. 7, pp 249 |
| **41.** | Input enhancement in string matching | Horspool algorithm | CO4 | T1: Ch. No. 7, pp 255 |
| **42.** | | Horspool algorithm | CO4 | T1: Ch. No. 7, pp 255 |
| **43.** | | Boyer-Moore algorithm | CO4 | T1: Ch. No. 7, pp 259 |
| **44.** | | Boyer-Moore algorithm | CO4 | T1: Ch. No. 7, pp 259 |
| **45.** | Limitations of algorithm power | Introduction, | CO4 | T1: Ch. No. 11, pp 379 |
| **46.** | | lower bound arguments | CO4 | T1: Ch. No. 11, pp 380 |
| **47.** | | decision trees | CO4 | T1: Ch. No. 11, pp 386 |
| **48.** | | decision trees | CO4 | T1: Ch. No. 11, pp 386 |
| **49.** | | P, NP and NP-complete problems | CO4 | T1: Ch. No. 11, pp 393 |
| **50.** | | P, NP and NP-complete problems | CO4 | T1: Ch. No. 11, pp 393 |
| **Module 5** | | | | |
| **51.** | Coping with limitations of algorithm power | Introduction | CO5 | T1: Ch. No. 12, pp 415 |
| **52.** | Backtracking | n-queens problem | CO5 | T1: Ch. No. 12, pp 417 |
| **53.** | | subset-sum problem | CO5 | T1: Ch. No. 12, pp 419 |
| **54.** | | Hamiltonian circuit problem | CO5 | T1: Ch. No. 12, pp 418 |
| **55.** | Branch & bound technique | Introduction | CO5 | T1: Ch. No. 12, pp 424 |
| **56.** | | Travelling salesperson problem | CO5 | T1: Ch. No. 12, pp 430 |
| **57.** | | Travelling salesperson problem | CO5 | T1: Ch. No. 12, pp 430 |
| **58.** | | knapsack problem | CO5 | T1: Ch. No. 12, pp 428 |
| **59.** | | knapsack problem | CO5 | T1: Ch. No. 12, pp 428 |
| **60.** | | Job assignment problem | CO5 | T1: Ch. No. 12, pp 425 |
| **61.** | | Job assignment problem | CO5 | T1: Ch. No. 12, pp 425 |

**The main pedagogical methods in the course are as follows:**

- Lecture mode.
- Power Point Presentation.
- Seminar by students.
- Video based learning.
- Problem based learning method.

- Simulation Practical system case study/Model Design.

| TABLE 4: SPECIAL DELIVERY METHOD | | | |
|---|---|---|---|
| **S. No** | **Session Number** | **Subtopic (as per lesson plan)** | **Pedagogical Method** |
| **1.** | 22 | Quick sort | Problem based learning method |
| **2.** | 41 | P, NP and NP-complete problems | Power Point Presentation |
| **3.** | 59 | Job assignment problem | Video based learning |

## 8. ASSESSMENT SCHEDULE

| TABLE 5: ASSESSMENT SCHEDULE | | | | | | |
|---|---|---|---|---|---|---|
| **Sl. No** | **Assessment Type** | **Coverage** | **CO Number(s)** | **Duration in Minutes** | **Marks** | **Weightage** |
| **1.** | Class Test-1 (CA1) | Module-1 & Half of Module-2 | CO1 & CO2 | 40 | 20 | 10% |
| **2.** | Midterm Exam | Module- 1,2 & Half of Module-3 | CO1, CO2 & CO3 | 90 | 50 | 25% |
| **3.** | Class Test-2 (CA2) | Second half of Module- 3& Module 4 | CO3 & CO4 | 40 | 20 | 10% |
| **4.** | CA3 (Assignment) | Module-2, 3, 4, & 5 | CO2, CO3, CO4 & CO5 | NA | 10 | 5% |
| **5.** | End Term Examination | Module-1,2,3 4 & 5 | CO1, CO2, CO3, CO4 & CO5 | 180 | 100 | 50% |

## 9. COURSE CLEARANCE CRITERIA:

This is in accordance with the Academic Regulations of the University and the Program Regulations and Curriculum of the respective program.

## 10. SAMPLE QUESTIONS:

| TABLE 6: SAMPLE QUESTIONS | | | | |
|---|---|---|---|---|
| **Sl. No** | **Question** | **Marks** | **CO Number** | **Blooms Cognitive Level** |
| 1 | Given an algorithm that recursively calls itself, discuss one method to determine its time complexity. | 10 | CO1 | **Apply** |
| 2 | Given a sorted array A, size N, of integers; Every element appears twice except for one. Demonstrate an algorithm to find that element in linear time complexity and without using extra memory. | 10 | CO2 | **Apply** |

## 11. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

| TABLE 7: SDG MAPPING | | | |
|---|---|---|---|
| **S. No** | **Topic** | **SDG Number** | **Justification** |
| **1** | Quick sort | SDG 4 | Students understand divide-and-conquer strategy and algorithmic efficiency by fostering structured thinking and strong foundational learning in computer science. |
| **3** | Job Assignment Problem | SDG 8 | enable fair and efficient allocation of resources and workforce by improving productivity and employment efficiency. |

## 12. CRITERIA FOR COURSE OUTCOME ATTAINMENT CALCULATION:

| TABLE 8: Threshold and Target Set for Course Outcomes | | | | |
|---|---|---|---|---|
| **Sl. No** | **C.O. No.** | **Course Outcomes** | **Threshold in %** | **Target in %** |
| 1. | CO1 | Compute efficiency of a given algorithm | 65 | 70 |
| 2. | CO2 | Apply divide and conquer technique for searching and sorting Problems | 55 | 60 |
| 3. | CO3 | Apply the Dynamic Programming technique for a given problem | 60 | 65 |
| 4. | CO4 | Apply greedy technique for solving a Problem | 55 | 60 |
| 5. | CO5 | Demonstrate Back tracking technique and limitations of Algorithms | 60 | 65 |

## 13. SUMMARY:

| TABLE 9: SUMMARY OF COURSE SCHEDULE | | | | |
|---|---|---|---|---|
| **Sl. No.** | **Activity** | **Start date** | **End date** | **Total number of Sessions** |
| **1.** | Program Integration & Over View of the course | 7-1-2026 | 7-1-2026 | **1** |
| **2.** | **Module : 01:** Introduction, Fundamentals of analysis of algorithm efficiency, Brute Force technique | 8-1-2026 | 2-2-2026 | **14** |
| **3.** | **Module : 02:** Divide & Conquer technique, Decrease & Conquer technique | 3-2-2026 | 19-2-2026 | **10** |
| **4.** | Assignment- I (Test-1) | | | |
| **5.** | **Module:03:** Dynamic Programming, Greedy method | 20-2-2026 | 20-3-2026 | **9** |
| **6.** | **Midterm Exam** | 10-3-2026 | 14-3-2026 | |

| | | | | |
|---|---|---|---|---|
| **7.** | Midterm Exam question paper discussion | 16-3-2026 | 16-3-2026 | **1** |
| **8.** | Assignment- II (Test-2) | | | |
| **9.** | **Module:04:** Space and time tradeoff, Limitations of algorithm power | 21-3-2026 | 10-4-2026 | **10** |
| **10.** | **Module:05:** Coping with limitations of algorithm power, Branch & bound | 13-4-2026 | 5-5-2026 | **11** |

**CONTACT TIMINGS IN THE CHAMBER FOR DISCUSSION**

Students can meet the respective course instructor during the Chamber Consultation Hour to clarify doubts related to the course.

**SPECIFIC GUIDELINES TO STUDENTS, IF ANY:**

- Attend all classes regularly.
- Bring a scientific calculator to every class.
- Refer to online study materials and watch the suggested videos available on the NPTEL website.

Dr Madhusudhan M V



Name and Signature of the course In-Charge

APPROVAL:

This course has been duly verified and approved by the Departmental Academic Committee (DAC).

Name and Signature of the Chairperson - DAC